

DS HOMEWORK Exercise no-3

1.write a c program to store n elements in array using dynamic memeory allocation and find the largest member?

```
#include <stdio.h>
#include<stdlib.h>
int main()
{
    int n,i;
    int *arr;
    printf("\nEnter n=");
    scanf("%d",&n);
    arr=(int*)malloc(n*sizeof(int));
    printf("\nEnter elements=");
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int large=arr[0];
    for(i=1;i<n;i++)
    {
        if(large<arr[i])
        {
            large=arr[i];
        }
    }
```

```

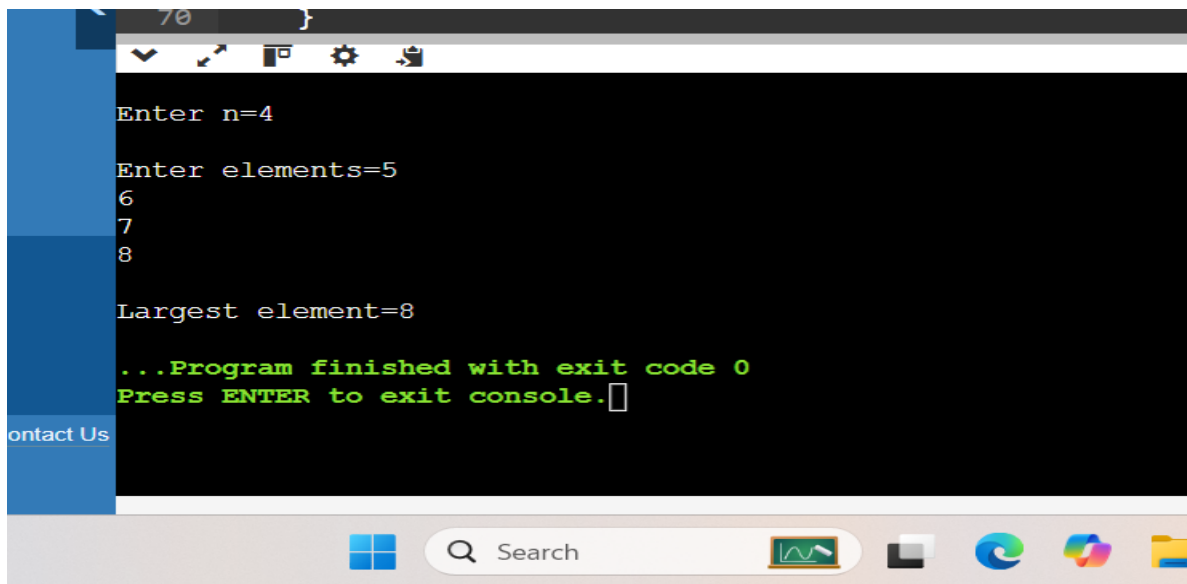
    }

    printf("\nLargest element=%d",large);

    free(arr);

    return 0;
}

```



```

Enter n=4
Enter elements=5
6
7
8

Largest element=8

...Program finished with exit code 0
Press ENTER to exit console.

```

2.Print an array in reverse order using pointer?

```

#include <stdio.h>
#include<stdlib.h>

int main()
{
    int n,i;
    int *arr;
    printf("\nEnter n=");
    scanf("%d",&n);

```

```

arr=(int*)malloc(n*sizeof(int));
printf("\nEnter elements=");
for(i=0;i<n;i++)
{
    scanf("%d",&arr[i]);
}

int *ptr=arr;
printf("\nreverse order=");
for(i=n-1;i>=0;i--)
{
    printf("%d",*(ptr+i));
}
free(arr);
return 0;
}

```



```

input
Enter n=4
Enter elements=5
6
7
8
5
reverse order=8765

```

3.write a c program to store the information of student using structure and sort them by names?

```
#include<stdio.h>
```

```

#include<string.h>

struct student
{
    int rollno;
    char name[20];
    float mark;
};

int main()
{
    int n,i,j;
    printf("Enter n=");
    scanf("%d",&n);
    struct student s[n];
    for(i=0;i<n;i++)
    {
        printf("\nEnter roll no= ");
        scanf("%d",&s[i].rollno);
        printf("\nEnter Your name:");
        scanf("%s",s[i].name)
        printf("\nEnter marks=");
        scanf("%f",&s[i].mark);
    }
    struct student temp;

    for(i=0;i<n;i++)

```

```

{
    for(j=i+1;j<n;j++)
    {
        if(strcmp(s[i].name,s[j].name)>0)
        {
            temp=s[j];
            s[j]=s[i];
            s[i]=temp;
        }
    }
}

for(i=0;i<n;i++)
{
    printf("\n-----Detail of student %d",i+1);
    printf("\nRoll no= %d ",s[i].rollno);
    printf("\nName:%s",s[i].name);
    printf("\nMarks=%.2f",s[i].mark);
}

return 0;
}

```

```
input
Enter n=3

Enter roll no= 101

Enter Your name:yuva

Enter marks=334

Enter roll no= 103

Enter Your name:asmi

Enter marks=67

Enter roll no= 103

Enter Your name:hari

Enter marks=34

-----Detail of student 1
Roll no= 103
Name:asmi
Marks=67.00
-----Detail of student 2
Roll no= 103
Name:hari
Marks=34.00
-----Detail of student 3
Roll no= 101
Name:yuva
Marks=334.00

...Program finished with exit code 0
Press ENTER to exit console.
```

4.write a c program to add n distance given in inch-feet system using structure?

```
#include<stdio.h>
```

```
struct distance
```

```
{
```

```
    int inch;
```

```
    int feet;
```

```

};

int main()
{
    int n,i;
    printf("\nEnter n=");
    scanf("%d",&n);
    struct distance d[n];
    printf("\nEnter distance");
    for(i=0;i<n;i++)
    {
        printf("\ninch=");
        scanf("%d",&d[i].inch);
        printf("\nFeet=");
        scanf("%d",&d[i].feet);
    }
    struct distance total;
    total.inch=0;
    total.feet=0;
    printf("\nTotal distance");
    for(i=0;i<n;i++)
    {
        total.inch+=d[i].inch;
        total.feet+=d[i].feet;
    }
    while(total.feet>=12)

```

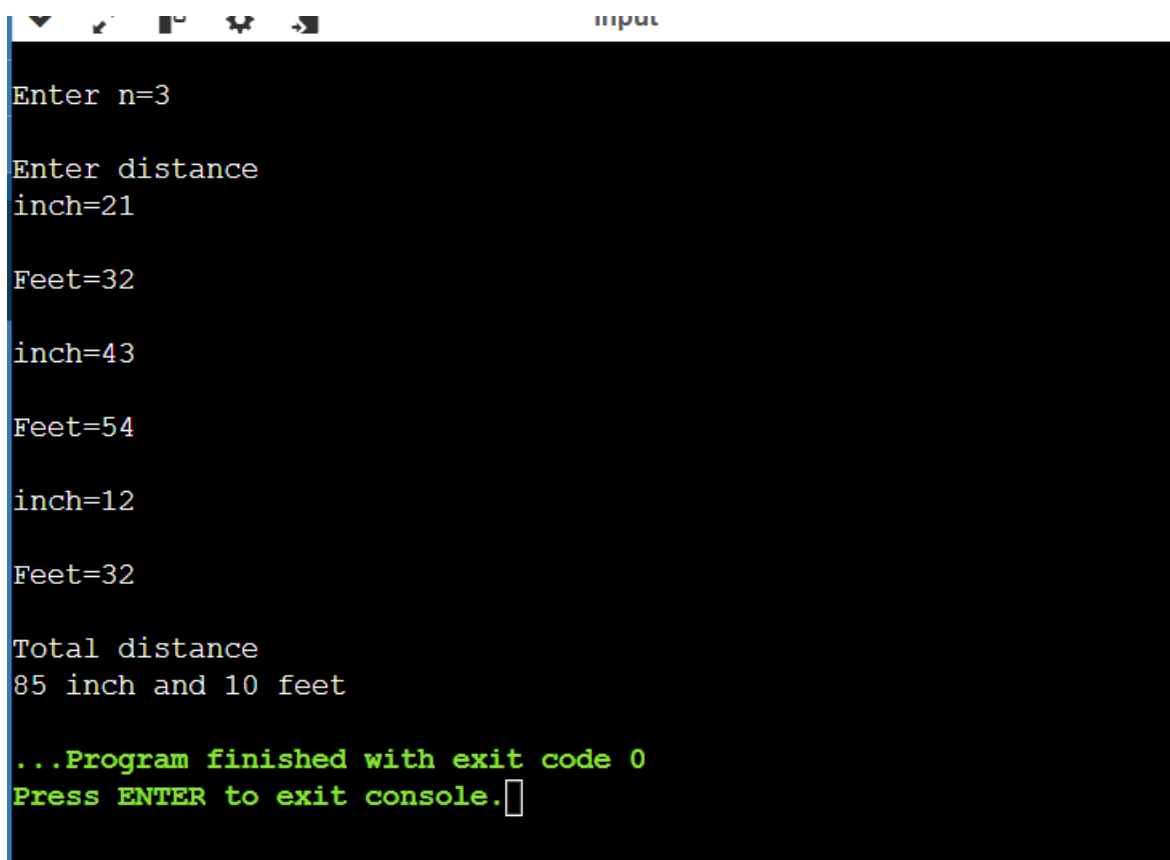
```

{
    total.inch++;
    total.feet-=12;
}

printf("\n%d inch and %d feet",total.inch,total.feet);

}

```



```

input
Enter n=3
Enter distance
inch=21
Feet=32
inch=43
Feet=54
inch=12
Feet=32
Total distance
85 inch and 10 feet
...Program finished with exit code 0
Press ENTER to exit console.

```

5. Write a C program to add two complex numbers by passing structure to the function?

```
#include<stdio.h>
```

```
struct complex
```

```
{
```

```
    int real;
```

```
    int img;
```



```
};
```

```
int main()
```

```
{
```

```
    struct complex add(struct complex c1 , struct complex c2);
```

```
    struct complex c1,c2;
```

```
    printf("\nEnter complex number 1");
```

```
    printf("\nreal=");
```

```
    scanf("%d",&c1.real);
```

```
    printf("\nimg=");
```

```
    scanf("%d",&c1.img);
```

```
    printf("\nEnter complex number 2");
```

```
    printf("\nreal=");
```

```
    scanf("%d",&c2.real);
```

```
    printf("\nimg=");
```

```
    scanf("%d",&c2.img);
```

```
    struct complex ans;
```

```
    ans=add(c1,c2);
```

```
    printf("\n%d+i%d",ans.real,ans.img);
```

```
    return 0;
```

```
}
```

```
struct complex add(struct complex c1, struct complex c2)
```

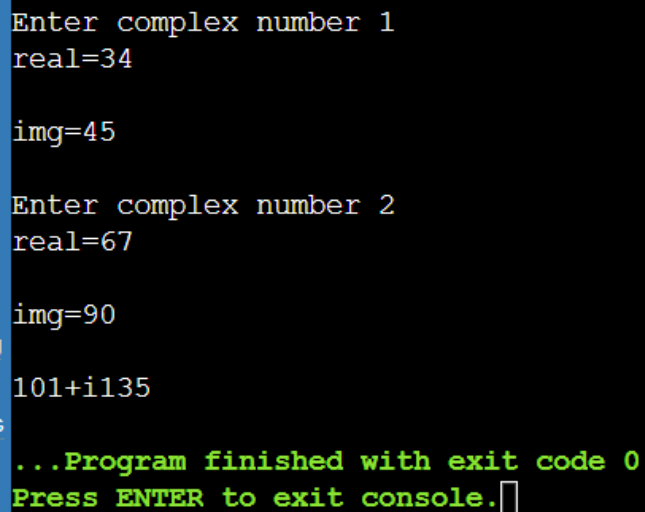
```
{
```

```
    struct complex temp;
```

```
    temp.real=c1.real+c2.real;
```

```
    temp.img=c1.img+c2.img;
```

```
    return temp;
}
```

A screenshot of a terminal window with a black background and green text. It shows the execution of a C program for complex numbers. The program prompts for two complex numbers. The first number has a real part of 34 and an imaginary part of 45. The second number has a real part of 67 and an imaginary part of 90. The program then outputs the sum of these two numbers as 101+i135. Finally, it displays a green message: "...Program finished with exit code 0" and "Press ENTER to exit console." with a cursor at the end.

```
Enter complex number 1
real=34
img=45
Enter complex number 2
real=67
img=90
101+i135
...Program finished with exit code 0
Press ENTER to exit console.
```

6.write a c program to maintain inventory management system?

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
struct inventory
```

```
{
```

```
    int prod_id;
```

```
    char prod_name[20];
```

```
    int quantity;
```

```
    float price;
```

```
};
```

```
int main()
```

```
{
```

```
    int n,i;
```

```
    printf("\nEnter no.of.product=");
```

```
    scanf("%d",&n);
```

```

struct inventory *inv;
inv=(struct inventory*)malloc(n*sizeof(struct inventory));
for(i=0;i<n;i++)
{
    printf("\nEnter product id=");
    scanf("%d",&inv[i].prod_id);
    printf("\nEnter product name:");
    scanf("%s",inv[i].prod_name);
    printf("\nEnter quantity=");
    scanf("%d",&inv[i].quantity);
    printf("\nEnter Product price=");
    scanf("%f",&inv[i].price);

}

int choice;
do{
    printf("\n1-add product\n2-display product\n3-search product\n4-delete
product\n5-update quantity\n6-exit");
    printf("\nEnter your choice=");
    scanf("%d",&choice);

    switch(choice)
    {
        case 1:
            int m;

```

```

printf("\nEnter no.of.product to add=");
scanf("%d",&m);

inv=(struct inventory*)realloc(inv,(m+n)*sizeof(struct inventory));

for(i=n;i<m+n;i++)
{

printf("\nEnter product id=");
scanf("%d",&inv[i].prod_id);
printf("\nEnter product name=");
scanf("%s",inv[i].prod_name);
printf("\nEnter quantity=");
scanf("%d",&inv[i].quantity);
printf("\nEnter Product price=");
scanf("%f",&inv[i].price);

}

n=n+m;

break;

case 2:
printf("\n-----");

```

```

printf("\n      inventory product      ");
printf("\n-----");
printf("\npid\tpname\tpquantity\tpprice");
for(i=0;i<n;i++)
{
    printf("\n%d\t                                     %-
10s\t%d\t%.2f",inv[i].prod_id,inv[i].prod_name,inv[i].quantity,inv[i].price);
}
break;

case 3:
int id,found=0,pos=-1;
printf("\nenter te product to be searched");
scanf("%d",&id);
for(i=0;i<n;i++)
{
    if(id==inv[i].prod_id)
    {
        found=1;
        pos=i;
    }
}
if(found==1)
{
    printf("\nptoduct found\n");
    printf("\npid\tpname\tpquantity\tpprice");
}

```

[illegible]

```
}
else
{
    printf("\nnot found");
}
break;
```

```
case 4:  
printf("\nEnter product id to be deleted=");  
scanf("%d",&id);
```

```
int posdel;
for(i=0;i<n;i++)
{
    if(id==inv[i].prod_id)
    {

        posdel=i;
    }
}
```

```

    }
    for(i=posdel;i<n;i++)
    {
        inv[i]=inv[i+1];
    }
    n--;

break;
case 5:
int qua;
printf("\nEnter product id for which quantity to be updated");
scanf("%d",&id);
printf("\nEnter the updated quantity=");
scanf("%d",&qua);
int posqua;
    for(i=0;i<n;i++)
    {
        if(id==inv[i].prod_id)
        {

            posqua=i;
        }
    }

    inv[posqua].quantity=qua;
break;

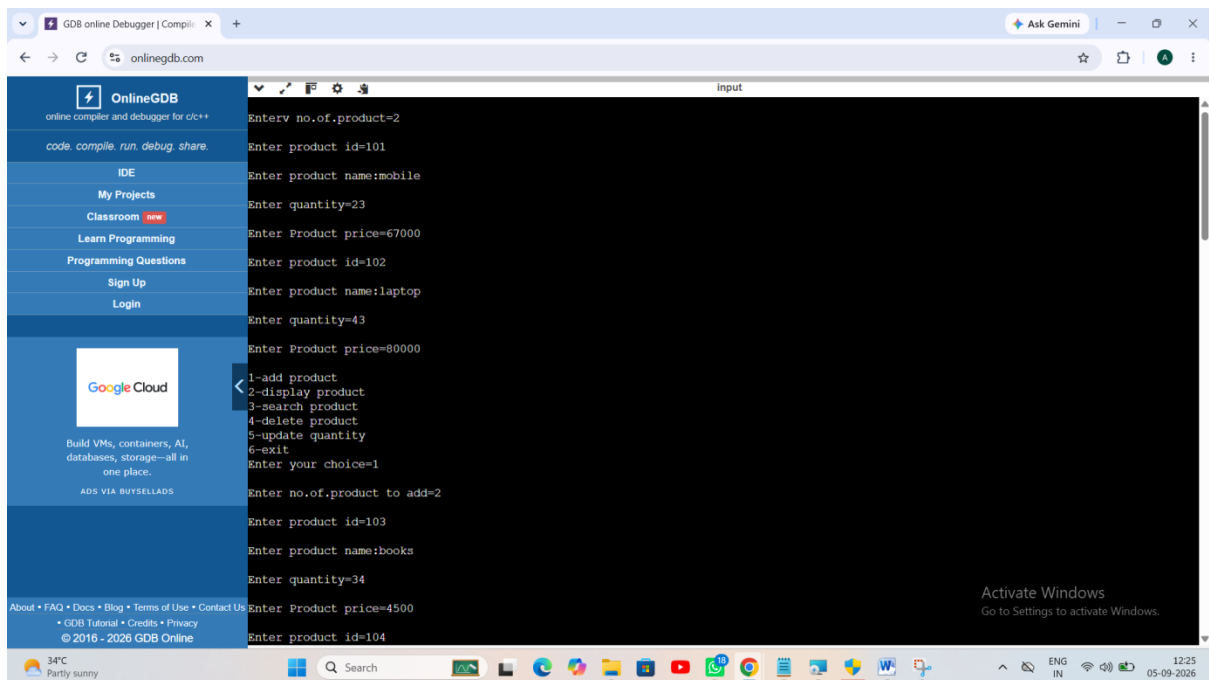
```

```

        case 6:
            printf("\nExit");
            break;

    }
}
while(choice<6);
free(inv);
return 0;
}

```



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input

```

Enter product id=104
Enter product name=headphone
Enter quantity=45
Enter Product price=7800
1-add product
2-display product
3-search product
4-delete product
5-update quantity
6-exit
Enter your choice=2

inventory product
-----
pid  pname  pquantity  pprice
101  mobile  23         67000.00
102  laptop  43         80000.00
103  books   34         4500.00
104  headphone 45         7800.00
1-add product
2-display product
3-search product
4-delete product
5-update quantity
6-exit
Enter your choice=3
enter te product to be searched102
product found
pid  pname  pquantity  pprice
101  mobile  23         67000.00
102  laptop  43         80000.00
103  books   34         4500.00
104  headphone 45         7800.00
1-add product
2-display product
3-search product
4-delete product
5-update quantity
6-exit
Enter your choice=3

```

Snipping Tool

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Automatically saved to screenshots folder.

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input

```

Enter your choice=3
enter te product to be searched102
product found
pid  pname  pquantity  pprice
102  laptop  43         80000.00
1-add product
2-display product
3-search product
4-delete product
5-update quantity
6-exit
Enter your choice=4
Enter product id to be deleted=104
1-add product
2-display product
3-search product
4-delete product
5-update quantity
6-exit
Enter your choice=2

inventory product
-----
pid  pname  pquantity  pprice
101  mobile  23         67000.00
102  laptop  43         80000.00
103  books   34         4500.00
1-add product
2-display product
3-search product
4-delete product
5-update quantity
6-exit

```

Activate Windows

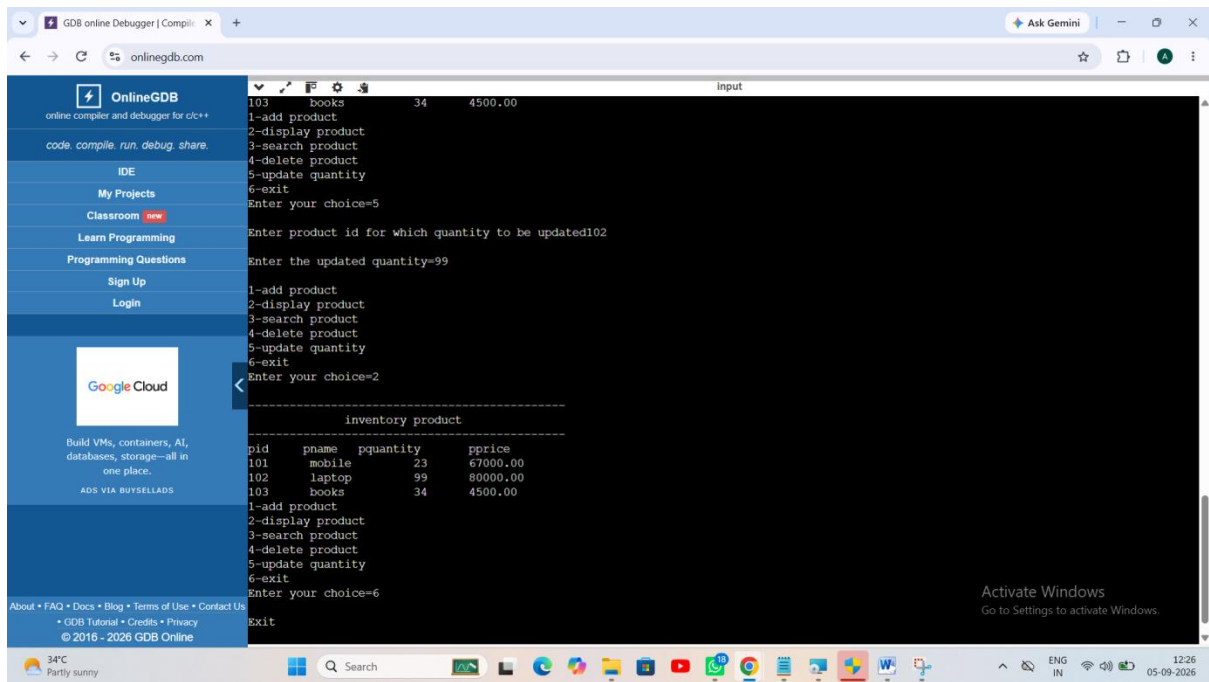
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7. Write a C program for decimal to binary, octal and hexadecimal conversion.

```
#include<stdio.h>
```

```
#define max 100
```

```
int top=-1;
```

```
int stack[max];
```

```
void push(int x)
```

```
{
```

```
    top++;
```

```
    stack[top]=x;
```

```
}
```

```
int pop()
```

```
{
```

```
    int val=stack[top];
```

```
    top--;  
    return val;  
}
```

```
int main()  
{  
    int num,temp;  
    printf("\nEnter decimal number=");  
    scanf("%d",&num);  
    temp=num;  
    while(num>0)  
    {  
        push(num%2);  
        num=num/2;  
    }  
    printf("\nDecimal to binary=");  
    while(top!=-1)  
    {  
        printf("%d",pop(num));  
    }  
  
    num=temp;  
    while(num>0)  
    {  
        push(num%8);
```

```

        num=num/8;

    }
    printf("\nDecimal to octat=");
    while(top!=-1)
    {
        printf("%d",pop());
    }

    num=temp;
    while(num>0)
    {
        push(num%16);
        num=num/16;

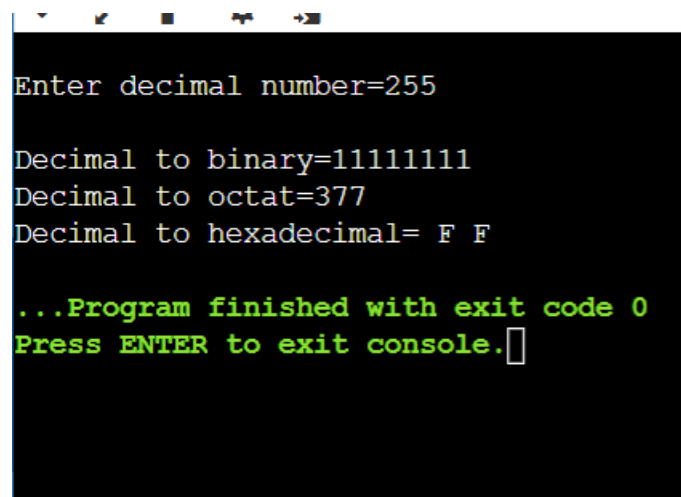
    }
    printf("\nDecimal to hexadecimal=");
    while(top!=-1)
    {
        int x=pop();
        if(x<10)
        {
            printf("%d",x);
        }
        else
        {

```

```

        printf(" %c",x+55);
    }
}
return 0;
}

```



```

Enter decimal number=255

Decimal to binary=11111111
Decimal to octat=377
Decimal to hexadecimal= F F

...Program finished with exit code 0
Press ENTER to exit console.

```

8. write a c program to calculate difference between two time period.

```

#include<stdio.h>

struct times
{
    int hour;
    int minute;
    int second;
};

int main()
{
    struct times t1,t2;
    printf("\nEnter time 1");
}

```

```
printf("\nHours=");
scanf("%d",&t1.hour);
printf("\nMinutes=");
scanf("%d",&t1.minute);
printf("\nSeconds=");
scanf("%d",&t1.second);
printf("\nEnter time 2");
printf("\nHours=");
scanf("%d",&t2.hour);
printf("\nMinutes=");
scanf("%d",&t2.minute);
printf("\nSeconds=");
scanf("%d",&t2.second);
```

```
struct times diff;
int total1=0,total2=0;
total1=t1.hour*3600+t1.minute*60+t1.second;
total2=t2.hour*3600+t2.minute*60+t2.second;
if(total1>total2)
{
    diff.second=total1-total2;
}
else
{
    diff.second=total2-total1;
```

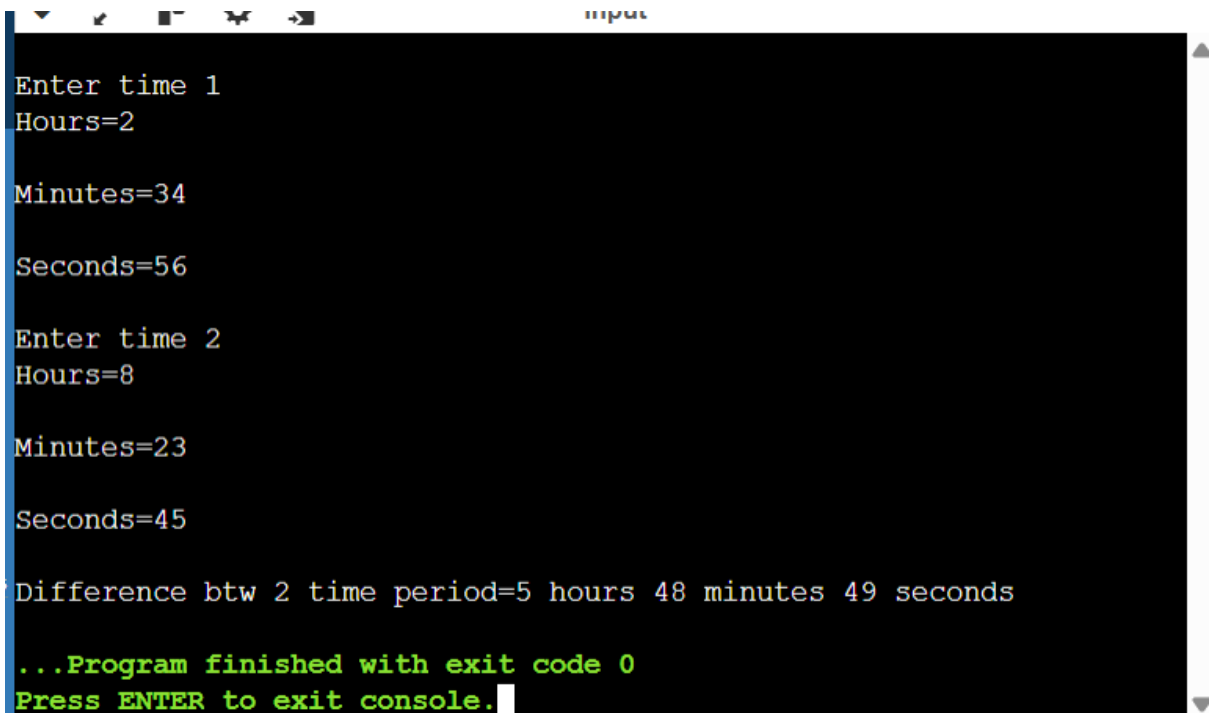
```

    }

    diff.hour=diff.second/3600;
    diff.second=diff.second%3600;
    diff.minute=diff.second/60;
    diff.second=diff.second%60;

    printf("\nDifference btw 2 time period=%d hours %d minutes %d
seconds",diff.hour,diff.minute,diff.second);
return 0;
}

```



```

input
Enter time 1
Hours=2
Minutes=34
Seconds=56
Enter time 2
Hours=8
Minutes=23
Seconds=45
Difference btw 2 time period=5 hours 48 minutes 49 seconds
...Program finished with exit code 0
Press ENTER to exit console.

```

9.write a c program to implement employee management system using structure.

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
struct employee
```

```

{
    int emp_id;
    char emp_name[20];
    char dept[20];
    int salary;
};

int main()
{
    int n,i;
    printf("\nEnter no.of.employee=");
    scanf("%d",&n);
    struct employee *e;
    e=(struct employee*)malloc(n*sizeof(struct employee));
    for(i=0;i<n;i++)
    {
        printf("\nEnter employee id=");
        scanf("%d",&e[i].emp_id);
        printf("\nEnter employee name:");
        scanf("%s",e[i].emp_name);
        printf("\nEnter department=");
        scanf("%s",e[i].dept);
        printf("\nEnter employee salary=");
        scanf("%d",&e[i].salary);
    }
}

```



```

int choice;

do{

    printf("\n 1-add employee\n2-display employee\n3-search employee\n4-
count employee\n5-delete employee\n6-exit");

    printf("\nEnter your choice=");

    scanf("%d",&choice);


    switch(choice)
    {

        case 1:

            int m;

            printf("\nEnter no.of.employee to add=");

            scanf("%d",&m);


            e=(struct employee*)realloc(e,(m+n)*sizeof(struct employee));


            for(i=n;i<m+n;i++)
            {

                printf("\nEnter employee id=");

                scanf("%d",&e[i].emp_id);

                printf("\nEnter employee name=");

                scanf("%s",e[i].emp_name);

                printf("\nEnter department=");

                scanf("%s",e[i].dept);

```

```

        printf("\nEnter employee salary=");
        scanf("%d",&e[i].salary);
    }
    n=n+m;

    break;

    case 2:

        printf("\n-----");
        printf("\n      employee details      ");
        printf("\n-----");
        printf("\nempid\temp_name\tedepartment\tesalary");
        for(i=0;i<n;i++)
        {
            printf("\n%d\t                %-10s\t%-
10s\t%d",e[i].emp_id,e[i].emp_name,e[i].dept,e[i].salary);
        }
        break;

        case 3:

        int id,found=0,pos=-1;
        printf("\nEnter the employee id to be searched");
        scanf("%d",&id);
        for(i=0;i<n;i++)
        {
            if(id==e[i].emp_id)
            {

```

```
        found=1;

        pos=i;
    }
}

if(found==1)
{
    printf("\nemployee found\n");


    printf("\nnempid\temp_name\tedepartment\tesalary");

    printf("\n%d\t                                     %-10s\t%-10s\t%d",e[pos].emp_id,e[pos].emp_name,e[pos].dept,e[pos].salary);


}

else

{
    printf("\nnot found");
}

break;


case 4:

printf("\nNo.of.employee=%d",n);

break;


case 5:

printf("\nEnter employee id to be deleted=");
```

```
scanf("%d",&id);

int empdel;
for(i=0;i<n;i++)
{
    if(id==e[i].emp_id)
    {

        empdel=i;
    }
}
for(i=empdel;i<n;i++)
{
    e[i]=e[i+1];
}
n--;

break;
case 6:
printf("Exit");
break;

}
```

```

}

while(choice<6);

return 0;

}

```

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```

Enter no.of.employee=2
Enter employee id=201
Enter employee name=assmitha
Enter department=it
Enter employee salary=78000
Enter employee id=202
Enter employee name=moni
Enter department=csc
Enter employee salary=50000
1-add employee
2-display employee
3-search employee
4-count employee
5-delete employee
6-exit
Enter your choice=1
Enter no.of.employee to add=1
Enter employee id=203
Enter employee name=vivek
Enter department=it
Enter employee salary=45000

```

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```

1-add employee
2-display employee
3-search employee
4-count employee
5-delete employee
6-exit
Enter your choice=2
-----
employee details
empid  emp_name  edepartment  esalary
201    assmitha  it           78000
202    moni      csc          50000
203    vivek     it           45000
1-add employee
2-display employee
3-search employee
4-count employee
5-delete employee
6-exit
Enter your choice=3
enter te employee id to be searched201
employee found
empid  emp_name  edepartment  esalary
201    assmitha  it           78000
1-add employee
2-display employee
3-search employee
4-count employee
5-delete employee
6-exit
Enter your choice=4

```

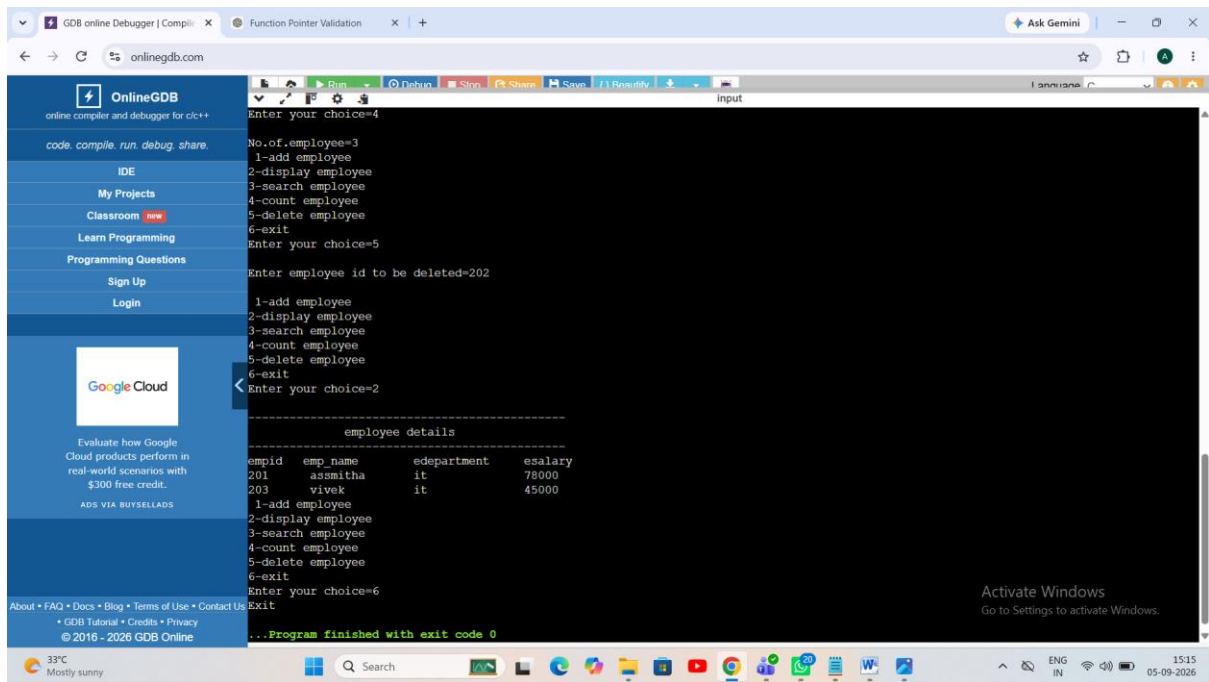
Activate Windows
Go to Settings to activate Windows.

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ENG IN

15:15 05-09-2026



10.write a c program to calculate the area and perimeter of triangle and rectangle using function pointer

```
#include<stdio.h>
```

```
void area_rec(int l,int b)
```

```
{ int area=0;
```

```
area=l*b;
```

```
printf("\nArea of rectangle=%d",area);
```

```
}
```

```
void area_tri(float l , float b)
```

```
{
```

```
float area= (0.5*l*b);
```

```
printf("\nArea of triangle=%.2f",area);
```

```
}
```

```

void perimeter_rec(int l , int b)
{

    int p= (2*l+2*b);
    printf("\nperimeter of rectangle=%d",p);
}

```

```

void perimeter_tri(int a , int b ,int c)
{
    int p= (a+b+c);
    printf("\nperimeter of triangle=%d",p);
}

```

```

int main()
{
    int l,b;
    printf("\nEnter length and breadth for rectangle=");
    scanf("%d%d",&l,&b);
    void(*fp1)(int ,int)=area_rec;
    fp1(l,b);
    fp1=perimeter_rec;
    fp1(l,b);
    float len,bre;
    printf("\nEnter length and breadth of triangle=");
    scanf("%f %f",&len,&bre);
}

```

```

void(*fp2)(float,float)=area_tri;
fp2(len,bre);

int p,q,r;
printf("\nEnter three sides of triangle=");
scanf("%d%d%d",&p,&q,&r);
void(*fp3)(int,int,int)=perimeter_tri;
fp3(p,q,r);
return 0;

}

```

```

Enter length and breadth for rectangle=4
7
Area of rectangle=28
perimeter of rectangle=22
Enter length and breadth of triangle=6.8
9.4
Area of triangle=31.96
Enter three sides of triangle=7
9
5
perimeter of triangle=21

...Program finished with exit code 0
Press ENTER to exit console.

```